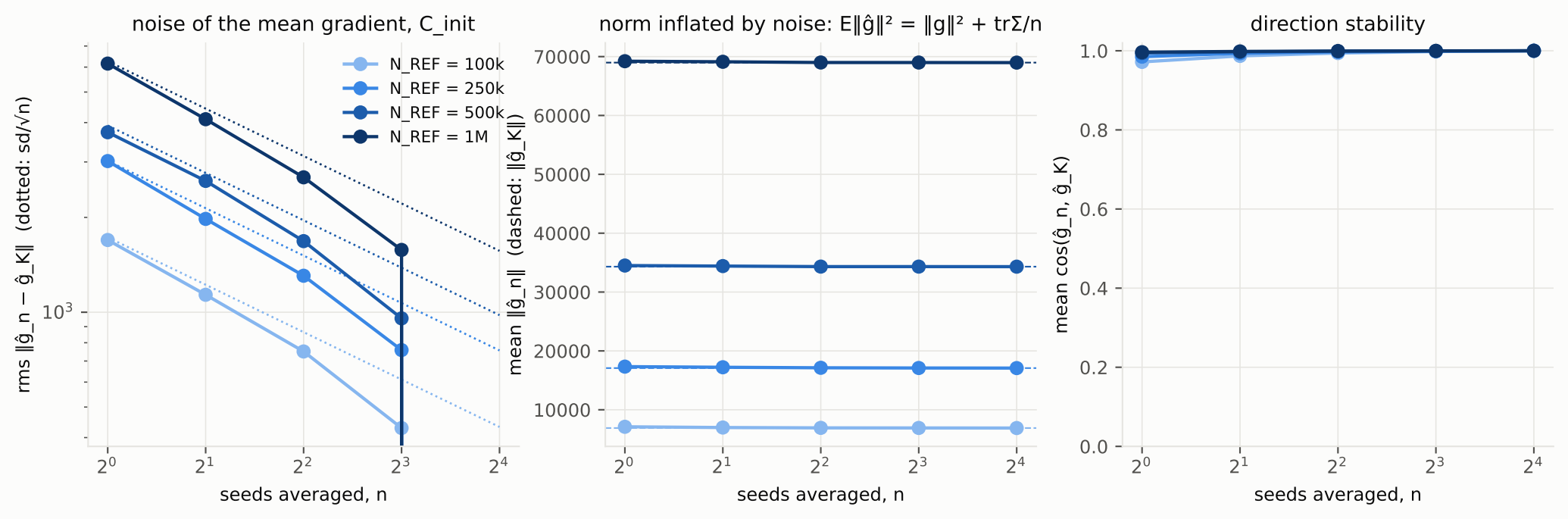
noise of the mean gradient, C_init
N_REF = 100k
N_REF = 250k
N_REF = 500k
N_REF = 1M
rms ‖ĝ_n − ĝ_K‖ (dotted: sd/√n)
10^3
2^0
2^1
2^2
2^3
2^4
seeds averaged, n
norm inflated by noise: E‖ĝ‖² = ‖g‖² + trΣ/n
mean ‖ĝ_n‖ (dashed: ‖ĝ_K‖)
70000
60000
50000
40000
30000
20000
10000
2^0
2^1
2^2
2^3
2^4
seeds averaged, n
direction stability
mean cos(ĝ_n, ĝ_K)
1.0
0.8
0.6
0.4
0.2
0.0
2^0
2^1
2^2
2^3
2^4
seeds averaged, n